

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester IV Examination, 2022-23

Zoology

Paper: ZOB-401 — Fundamental Endocrinology and Economic Zoology

Time: Three hours

Full Marks: 70

Note: Answer three questions from Section-A and three questions from Section-B. Question No. 1 in both sections is compulsory. Use a separate answer book for each section. The figures in the right-hand margin indicate marks.

SECTION – A
(Fundamental Endocrinology) (Marks: 35)

Question 1. Answer the following parts:

- (a) Define the following terms: [1 × 3 = 3]
- (i) Bursicon
 - (ii) Cretinism
 - (iii) Relaxin
- (b) Differentiate between the following: [2 × 2 = 4]
- (i) Negative and Positive feedback mechanisms
 - (ii) Sertoli cells and Leydig cells
- (c) Justify giving reasons whether the following statements are True or False: [2 × 2 = 4]
- (i) Environmental factors affect thyroid hormone secretions.
 - (ii) Glucagon is a hypoglycemic hormone.

Question 2. Answer the following parts:

- (a) Define hormones and classify them according to their structural and functional similarities. [6]
- (b) How does the pineal gland coordinate daily rhythms? Explain with the help of a flowchart. [6]

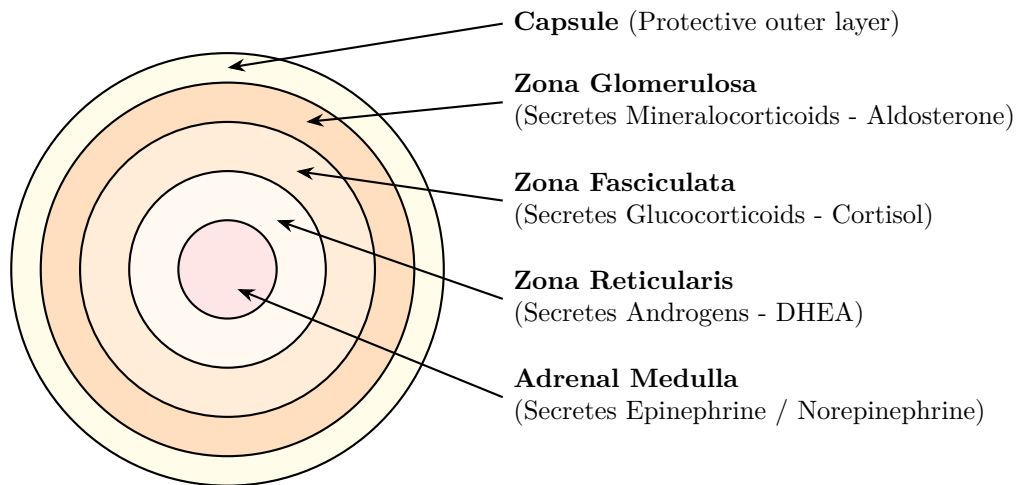
(Circadian Pacemaker);
ode (scg) at (0,-0.5) Superior Cervical Ganglion (SCG);
ode (pineal) at (0,-2) Pineal Gland;
ode (melatonin) at (0,-3.5) Melatonin Secretion
(High in Dark, Low in Light);
ode (rhythms) at (0,-5) [fill=green!10] Regulation of Circadian Rhythms

(Sleep-wake cycle, body temp);

[->, thick] (light) – (retina); [->, thick] (retina) – node[draw=none, fill=none, right] Retino-hypothalamic Tract (scn); [->, thick] (scn) – (scg); [->, thick] (scg) – node[draw=none, fill=none, right] Sympathetic pathway (pineal); [->, thick] (pineal) – (melatonin); [->, thick] (melatonin) – (rhythms);

Question 3. Answer the following parts:

- (a) Draw a neat and well-labeled diagram of the adrenal gland. Also mention the biological actions of all the adrenocortical hormones. [8]



- (b) Describe the neurohypophysial hormone family. [4]

Question 4. Write short notes on any two of the following: [6 × 2 = 12]

- (a) Calcium homeostasis

(≈ 10 mg/dL);

ode (low) at (-3.5, 1.5) [draw, rounded corners, fill=red!5] Low Blood Ca^{2+} ;

ode (pth) at (-3.5, 3.2) [draw, rounded corners, fill=orange!5] Parathyroid Gland releases **PTH**;

ode (pth-act) at (-3.5, 4.8) [draw, rounded corners, fill=orange!10] Reabsorption from kidney
Ca release from bone
Absorption in intestine;

ode (high) at (3.5, 1.5) [draw, rounded corners, fill=blue!5] High Blood Ca^{2+} ;

ode (calcitonin) at (3.5, 3.2) [draw, rounded corners, fill=cyan!5] Thyroid Gland releases **Calcitonin**;

ode (calc-act) at (3.5, 4.8) [draw, rounded corners, fill=cyan!10] Deposition in bones
Excretion in kidney;

[->, thick] (normal) to[out=135, in=270] (low); [->, thick] (low) – (pth); [->, thick] (pth) – (pth-act); [->, thick] (pth-act) to[out=0, in=90] (normal);

[->, thick] (normal) to[out=45, in=270] (high); [->, thick] (high) – (calcitonin); [->, thick] (calcitonin) – (calc-act); [->, thick] (calc-act) to[out=180, in=90] (normal);

Question 1. Answer the following parts:

(a) Write whether the following statements are True or False. Justify your answers in 2-3 sentences each: [2 × 3 = 6]

- (i) Female silk moths survive about 3 years.
- (ii) Male and female lac insects reside in similar shaped resin cells.
- (iii) Leghorn, Minorca and Ancona are good egg layers.

(b) Define the following terms: [1 × 5 = 5]

- (i) Pullets
- (ii) Heifer
- (iii) Bee-economics
- (iv) Seed lac
- (v) Waggle dance

Question 2. Describe the steps of fish culture, mode of breeding and methods of fish preservation. [12]

Question 3. Write an essay on milk and milk products. [12]

Question 4. Answer the following parts:

- (a) Write an essay on the diseases of silkworms. [8]
- (b) Write about the economic importance of honeybees. [4]

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